

Semantic Communication for Collaborative Embodied Question Answering in Multi-Agent Systems

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Abstract—The advancement of Large Language Models (LLMs) has propelled the development of AI agents capable of complex tasks. This paper focuses on collaborative multi-agent systems for Embodied Question Answering (EQA) application, where multiple agents must perceive environments, reason, and communicate to answer human queries. The semantic communication is recognized as the critical paradigm for efficient inter-agents information exchange within such systems. To address key challenges, we propose a novel agentic semantic communication framework. This framework integrates an agent’s cognitive Knowledge Base (KB) into the semantic communication process, establishes a distributed coordination mechanism to maintain consistent group knowledge, and develops efficient methods for high-level semantic extraction and joint source-channel semantic encoding mechanisms tailored for EQA tasks. Simulation results demonstrate that our proposed mechanisms effectively ensure cognitive consistency across agents, enable robust and efficient agentic semantic information transmission under low signal-to-noise ratio conditions, showcasing significant performance gains over baseline approaches.

Index Terms—Semantic communication, multi-agent systems, knowledge base coordination, agentic communication

I. INTRODUCTION

The rapid development of Large Language Models (LLMs) is driving advances in Artificial Intelligence (AI) agent technology, with widespread application across diverse domains [1], [2]. An AI agent is a new form of intelligent agent that utilizes the reasoning abilities of pre-trained LLMs to comprehend human language, interact with the physical environment, and execute complex tasks [3]. The capabilities of a single agent are often limited by its narrow knowledge and operational scope. Therefore, by forming a group of multiple agents, the collective intelligence can be substantially

enhanced through group information sharing. Through inter-agent communication, such groups can adapt to dynamically environments and effectively address tasks across a wider geographical area [4], [5].

Effective collaboration in multi-agent systems depends on efficient information exchange. Semantic communication, characterized by its high efficiency and robustness against interference, serves as a critical enabler [6], [7]. Specifically, we consider a multi-agent system designed for collaborative Embodied Question Answering (EQA) tasks (as illustrated in Fig. 1). In the proposed framework, a human queries a personal agent regarding environmental information. This agent relays the question to an appropriate remote agent, which performs scene perception to fulfill the task. Then, the acquired information is subsequently feedback to the personal agent, thereby responding to the human query. This system holds broad application potential in fields such as smart homes, intelligent cities and security and healthcare services. Compared to passive traditional video surveillance systems, this multi-agent collaborative EQA system is capable of active perception, reasoning, and interaction. By processing tasks and transmitting semantic information, the transmitted data volume is thereby drastically reduced while the level of intelligence is simultaneously enhanced.

This paper specifically focus on the information exchange among agents within collaborative EQA multi-agent system. Firstly, semantic communication is not just a suitable but a fundamentally communication paradigm for enabling efficient communication in multi-agent systems, particularly for a complex task like EQA. For agents with reasoning capability, semantic communication enables agents to transmit the results of their reasoning, rather than the raw sensory data. Furthermore, as human language is the input and output of LLM, current agent group utilizes human language for information exchange [8], which is inefficient for Machine-to-Machine (M2M) communication. Semantic communication provides a more efficient and machine-optimized information exchange paradigm, eliminating the inherent ambiguity and redundancy of human language. Secondly, the cognitive knowledge base (KB) of agents aids both in executing tasks and processing information, requiring that the semantic communication system of the collaborative EQA agents should be redesigned to fully incorporate agent’s cognitive KB in semantic information exchange. Finally, current research in semantic communication primarily focuses on information transmission between a

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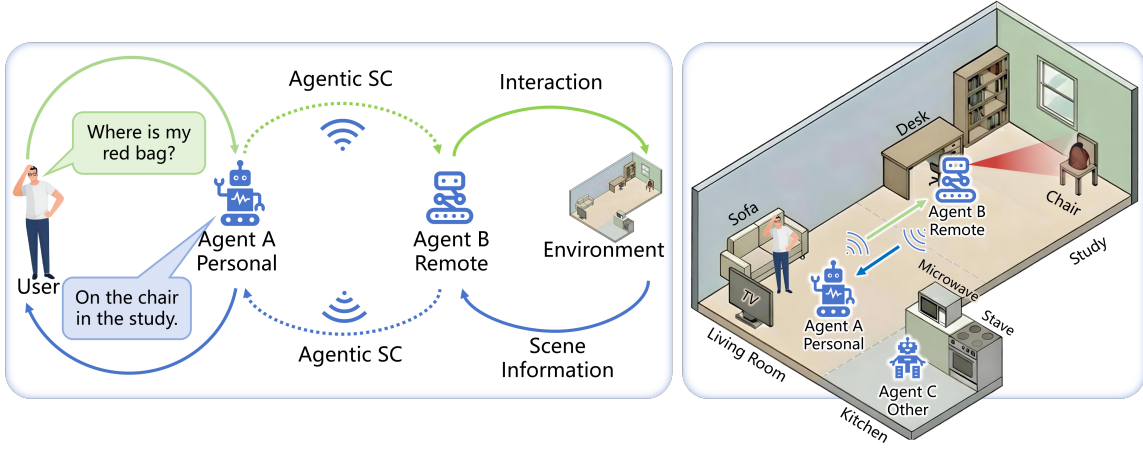


Fig. 1. The collaborative EQA multi-agent system.

single transmitter-receiver pair. Effective group-wide semantic understanding in multi-agent systems requires a consistent shared KB. Therefore, it is imperative to design a mechanism for the coordination of the group's semantic communication KB.

In view of the above requirements, the main contributions of our work are threefold. We especially determined the following three key issues for the collaborative EQA multi-agents system:

- **Communication paradigm.** To incorporate the agent cognitive KB into the semantic KB, we clarify the semantic KB with two components and refine the semantic information processing pipeline. Especially, the cognitive KB is represented by scene graph (in the form of Knowledge Graph (KG)), which facilitates the extraction of high-level semantic of multi-modal data, and reduces transmission load. This agentic semantic communication paradigm is fully described in Sec. III.
- **Communication network.** We design a distributed coordination mechanism for agents' cognitive KB to maintain unified and consistent knowledge information and facilitate effective multi-user semantic communication within the agent group (noted as DisBayesCKC). This mechanism is an essential underlying protocol for the effective networking of multi-agent clusters. The cognitive KB coordination mechanism is detailed in Sec. IV.
- **Communication content.** We develop mechanisms for high-level semantic information extraction and efficient semantic encoding, specifically tailored for EQA tasks. Based on the structured EQA cognitive KB, semantic information can be distilled into a compact KG form and further encoded efficiently through triplet entity embedding and compression techniques. The proposed triplet joint source and channel encoding mechanism is noted as Triplet-EmPress. The semantic information extraction and semantic encoding mechanism is discussed in Sec. V and Sec. VI.

The innovation of this paper lies in systematically introducing semantic communication into the practical task of

multi-agent EQA, and in proposing a semantic communication architecture centered on agents' cognitive knowledge base to address the information interaction requirements of agents in this task. The aforementioned research provides solutions to the core issues of collaborative EQA agentic semantic communication. Compared with existing work, the innovation of this paper is not merely a standalone improvement of a specific module or the proposal of an algorithm, but rather a system-level innovation in architectural design, mechanism integration, and application validation for real-world EQA scenarios, addressing the practical system-level challenges of semantic communication. Finally, we perform extensive simulations, and the results demonstrate that the proposed mechanisms effectively achieve cognitive consistency in the multi-agents system and efficiently transmit semantic information for EQA tasks under low signal-to-noise ratio conditions.

The rest of the paper is organized as follows. Related works are summarized in Sec. II. The agentic semantic communication, especially for EQA task, is fully introduced in Sec. III. Then, Sec. IV explains the cognitive KB coordination mechanism for the collaborative agent group. Sec. V details how to extract high-level semantic information in the form of triplets, and Sec. VI describes its source and channel encoding mechanisms. In Sec. VII, we evaluate the performance gain of our proposed agentic semantic communication mechanism on several baselines. Finally, we conclude this work in Sec. VIII.

II. RELATED WORKS

A. AI agent system

The development of AI agent systems has been shaped by the influences of expert systems, RL, and LLM. The early AI agent systems were rule-based, such as domain-specific expert systems [9]. These systems emphasized the importance of domain-specific knowledge, foreshadowing the application of structured KBs in modern AI agents. However, their ability to adapt to dynamic environments was limited. Subsequently, the rise of Deep Reinforcement Learning (DRL) [10] introduced autonomous decision-making capabilities, allowing agents to learn complex policies from high-dimensional input. Although

DRL excels in task-specific optimization, its reliance on predefined rewards restricts its adaptability in open-ended scenarios, such as those involving natural language understanding.

Recent studies have explored the application of LLM-driven AI agents in various domains. For example, in 6G networks, mobile AI agents leverage LLMs for network automation, personalized services, and multi-agent collaboration [11]. Similarly, WirelessAgent [12] leverages an LLM to understand the intent of the user and efficiently allocate resources, thus optimizing the slice of the wireless network. In the fields of IoT and smart homes, LLM agents such as SAGE [1] and CASIT [13] highlight the role of collective intelligence. SAGE uses dynamic action trees to manage device interactions and user preferences, while CASIT employs memory mechanisms to collaboratively process IoT data. These systems emphasize the importance of structured knowledge representation in AI agent systems, but do not address the extraction, compression and transmission of semantic information across agents. Our work bridges this gap.

B. Semantic Communication

Semantic communication enables wireless edge devices to extract and transmit the meaning of the original data (called semantic information) to improve the efficiency of wireless network communication. Recent advances in semantic communication emphasize the pivotal role of integrating KBs to enhance both communication efficiency and reliability. For example, in [14], the authors propose a framework that advocates a transition from data-driven AI-enhanced systems to reasoning-driven native AI networks, highlighting the need for compact and generalizable semantic representations.

With the development of LLMs, semantic communication is evolving towards generative semantic communication which also includes AI Agent communication [15], [16]. Its characteristic lies in using cognitive KB (in form of KG) to perform brain-like reasoning to achieve efficient utilization of semantics. In [17], authors introduces cognitive semantic communication, using external KGs to perform brain-like reasoning during transmission, thus improving spectral efficiency and communication reliability. In [18], the authors propose a generative semantic communication architecture with three sub-KBs (source, task, and channel) that enable efficient semantic encoding in low-dimensional subspaces. In [19], the authors advance the field by integrating KG for semantic error correction and reasoning, demonstrating superior compression performance and reliability in single-user and multi-user scenarios. In [20], the authors propose a KG-augmented semantic communication framework, in which KG context information is encoded in the original message and helps reconstruct the information in poor communication condition. KG-enabled semantic communication also shows advantages in unified semantic representation and alignment. By extracting cross-modal information into KGs, cross-modal semantic communication [21] ensures semantic consistency and inter-modal conversion.

Furthermore, KG-enhanced semantic communication often extracts the semantics of raw information into the form of

triples for transmission. In [22], the authors propose a probability graph structure for semantic communication, in which raw data is extracted into triplets, and the triplet with highest relation probability can be omitted and compressed during transmission. In [23], the authors propose KG-enhanced triplet semantic communication. The system leverages shared KG for semantic extraction, which is represented in the form of entity-relation-entity triplet. In [24], the authors present KGRAG-SC, a KG-assisted semantic framework that leverages semantic extraction and RAG for semantic transmission. These works leverage a shared KG to reduce data transmission but overlook two critical issues: how communicating parties maintain KG consistency and how to effectively encode and compress the triplets.

Synchronizing KBs is critical for maintaining semantic consistency in communication in distributed systems. In [25], the authors introduce a shared KB framework for text transmission, with the aim of reducing communication overhead by transmitting only residual information. However, the heterogeneity of KBs remains a challenge. In [26], the authors propose a joint distillation method to synchronize heterogeneous KBs, while [27] addresses this issue through synchronization signals and iterative optimization. In [28], the authors unify multi-task semantic communication through dynamic feature selection, optimizing transmission efficiency under various channel conditions. In [29], the authors develop a distributed knowledge base consensus scheme based on naming game to achieve mutual consensus of heterogeneous knowledge by a group of agents. Nevertheless, there is still limited research on the global consistency of the semantic communication cognitive KB within a group of intelligent agents.

Semantic communication serves as a force multiplier for agent information exchange. However, agentic semantic communication, particularly for multi-agent systems, remains in its infancy. It demands further research into multi-user KB coordination, semantic information extraction, and efficient transmission. Addressing this gap, our study investigates these challenges within a collaborative EQA scenario, presenting a novel framework for effective multi-agent semantic communication.

III. AGENTIC EQA SEMANTIC COMMUNICATION

A. Collaborative EQA Multi-agent System

The collaborative EQA multi-agent system is a framework where multiple autonomous AI agents, embodied within a shared physical environment, collaboratively perceive, reason, and act to answer human queries about the scene. As depicted in Fig. 1, when a user poses a question (e.g., “Where is my red bag?”) towards a personal agent, the agent forwards the query to other remote embodied agents (such as mobile robots or static sensors), which observe the physical environment and reply the answer (e.g., “Your red bag is on the chair in the study.”). Such system possesses broad application potential in fields including smart homes, intelligent urban systems, security, and healthcare services.

The core functionality of an EQA system is scene-based questioning and answering, where the dialogue inherently centers on the objects present in the environment. Hence, existing

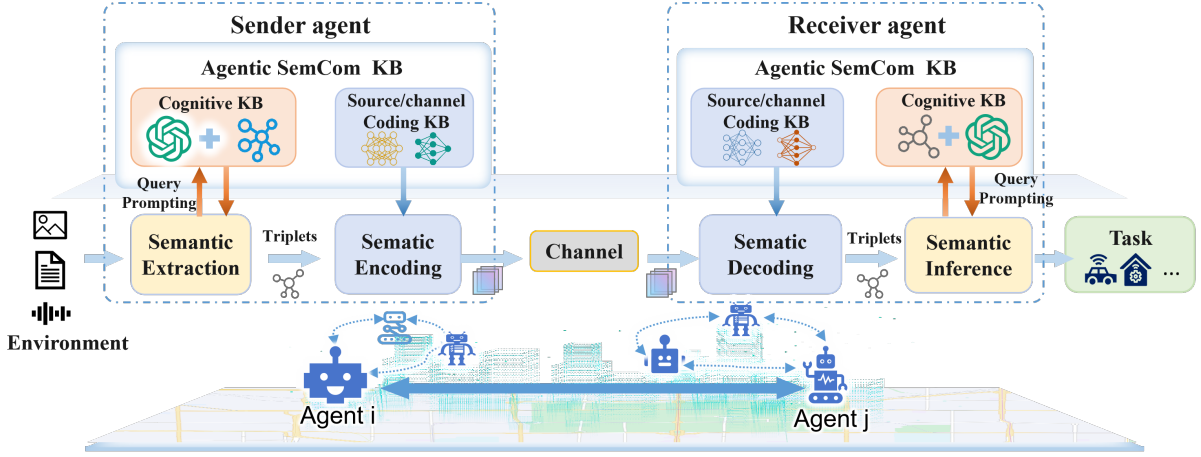


Fig. 2. Agentic semantic communication knowledge bases.

EQA agents are designed with an integrated prior cognitive KB of their environment, which is crucial for enabling accurate scene interpretation and responses [30], [31]. The focus of this paper is on inter-agent machine type communication. Leveraging the architectural features of EQA agents, we aim to design an agentic semantic communication system and its corresponding mechanisms tailored for such collaborative multi-agent EQA system.

B. SC in the Collaborative EQA system

Information exchange between AI agents primarily relies on human natural language. This is mainly because the core component of these agents—LLMs—processes and generates human language, and the design and ecosystem of AI agents are built around natural language prompts and interactions (e.g., LangChain, AutoGPT, etc.). However, communication between AI agents also belongs to the category of M2M communication. Human natural language, characterized by flexibility and redundancy, is not well-suited for machine communication scenarios. By adopting the paradigm of semantic communication, which involves structuring information into semantic representations and applying source coding compression with semantic encoding, the resource utilization efficiency and noise resilience of agent information transmission can be significantly enhanced.

As depicted in Fig. 2, based on the architecture of EQA agents and the semantic communication mechanism, this paper proposes the following agentic semantic communication framework and information process pipeline. First, the sender agent extracts multimodal information (such as visual, language) into high-level structured semantic information in the form of triplets. These triplets are then treated as the information source for transmission, undergoing source encoding and channel coding. At the receiving end, the receiver agent restores the received signals into triplets, which are subsequently used by the large model for downstream tasks such as task planning and action execution.

Compared to traditional semantic communication [32], the shared KB in an agentic semantic communication system

requires new dimensions. Firstly, for EQA agents, it is necessary to integrate a cognitive KB specifying the environment, leveraging technologies such as Retrieval-augmented Generation (RAG) to complete tasks like agent path planning, scene understanding and question reply. This cognitive KB is also beneficial for improving information transmission efficiency: during the process of extracting structured semantic information from raw multi-modal scene data, this KB helps identify the high-level semantic information that needs to be transmitted. Therefore, the cognitive KB shared by the collaborative multi-agent group serves as a crucial module for achieving semantic information extraction and understanding. Secondly, similar to traditional semantic communication mechanisms, the deep learning model that processes multimodal information into triples, as well as the deep learning model that encodes and compresses the triples to be transmitted, also constitute the shared KB for source encoding in agentic semantic communication.

C. EQA agent cognitive KB

LLMs possess general knowledge but lack a deep understanding of the specific and dynamically changing operational environments of EQA agents, such as spatial structures and their associated semantic information. By leveraging technologies like RAG, such unstructured scene information can be organized into a KB that agents can query accurately, in real time, and on demand. This enables the effective use of such knowledge in planning and decision-making for downstream tasks. *We refer to this type of KB, which supports EQA agent task execution, as the agent's cognitive KB.*

For an EQA agent, its cognitive knowledge primarily pertains to environmental information. Without loss of generality, environment can be represented in the form of a knowledge graph, which transforms complex unstructured environmental information into a machine-understandable and reasonable structured representation. Furthermore, for indoor scenes, their semantic information can be represented using a hierarchical scene graph, which is a typical structured representation method [33]. A hierarchical scene graph is a structured approach to representing scene information, that decomposes

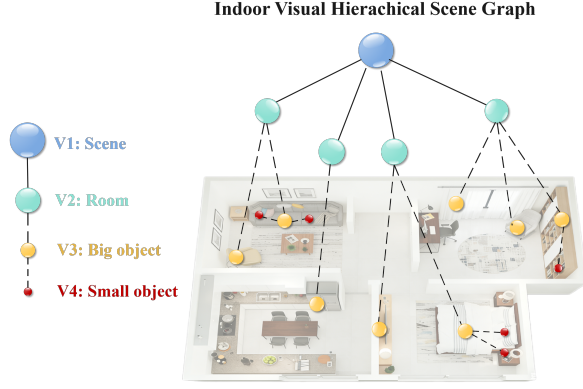


Fig. 3. Hierarchical scene graph.

and describes a complex scene from the macro to the micro level, from the overall to the local perspective. As illustrated in Fig. 3, an indoor space can be divided into the *Room* layer, the *Big_object* layer, and the *Small_object* layer. This hierarchical approach facilitates the agent’s understanding of object relationships in the environment and enables spatial comprehension and reasoning for answering questions. *Therefore, for an EQA agent, the representation form of its cognitive KB is the hierarchical scene graph.*

For information exchange among agents, the shared cognitive KB can significantly reduce the amount of data that needs to be transmitted. Similar to how humans use expressions like “the usual place” in communication, if information to be sent between agents exists in the cognitive KB, there is no need to explicitly transmit the full content—only the KB index needs to be sent. Then, the receiver can retrieve the complete information by querying the knowledge base. The cognitive KB serves as the shared memory and common knowledge for the agent group, and also represented as the world model for the operational environment of the EQA agent group. Therefore, for agentic semantic communication, the following three key issues need to be addressed:

- How can the agent group collectively maintain a unified and consistent cognitive KB such that they have the same shared cognitive knowledge?
- For EQA tasks, how can we extract the high-level semantics from questions and replies, and organize them into a hierarchical and structured framework to minimize transmission load by leveraging the cognitive KB?
- For semantic information that needs to be transmitted (in the form of triples), how can efficient semantic source-channel encoding and decoding be achieved?

The solutions to these three issues will be detailed in the subsequent three sections.

IV. AGENT GROUP COGNITIVE KNOWLEDGE COORDINATION

A. From Individual Cognition to Group Consensus

In collaborative EQA, an agent group must maintain a unified understanding of their shared environment to accurately and consistently reason about and respond to user

queries. This unified understanding is formalized as consistent cognitive KBs within the group. The cognitive KB serves as the agents’ structured internal representation of the environment, whose fundamental unit is a cognitive quadruplet. Let $\mathcal{Q} = \{q_1, q_2, \dots, q_M\}$ denote a cognitive KB, where each quadruplet is defined as $q_k = (u_k, \theta_k)$. Here, $u_k = (e_{1k}, r_k, e_{2k})$ is a semantic triplet representing a relationship between entities (e_{1k} and e_{2k} are entities, r_k describes their relationship). Let $\mathcal{U} = \{u_1, u_2, \dots, u_M\}$ denote the set of triples commonly concerned by all agents. We assume that each triplet is independent of each other and has a true degree of validity $\theta_k^* \in [0, 1]$ in the objective world. An agent’s cognition is thus characterized by its subjective belief, expressed as a probability distribution $p(\theta_k)$. Therefore, the cognitive KB is essentially the aggregation of all agents’ individual probabilistic beliefs. This section describes how a distributed coordination framework allows a multi-agent system to start with individual, potentially biased beliefs and ultimately form globally consistent and accurate group cognitive KBs.

Consider a multi-agent system for collaborative EQA with N agents distributed in a scene, denoted by $\mathcal{A} = \{a_1, a_i, \dots, a_N\}$. To preserve a measure of cognitive uncertainty in the estimation, we employ Bayesian estimation. Each agent a_i maintains, for each triplet u_k , a posterior probability distribution $p(\theta_{i,k})$ over the validity probability. Let $p(\theta_{i,k}^t)$ be agent a_i ’s posterior belief about θ_k^* at time step t . Our objective is to design a distributed collaboration protocol enabling the multi-agent system to achieve:

- 1) Accurate Estimation of validity probabilities:

$$\lim_{t \rightarrow \infty} \mathbb{E}[\theta_{i,k}^t] = \theta_k^*, \quad \forall a_i \in \mathcal{A}, u_k \in \mathcal{U}, \quad (1)$$

i.e., each agent’s expectation estimate for each triplet converges to the true probability.

- 2) Global Consensus on belief distributions:

$$\lim_{t \rightarrow \infty} p(\theta_{i,k}^t) = p(\theta_{j,k}^t), \quad \forall a_i, a_j \in \mathcal{A}, u_k \in \mathcal{U}, \quad (2)$$

i.e., the estimation distributions for any triplet converge across all agents. Please note that having agents share an identical estimate of expectation for a triplet does not constitute true consensus, as the posterior distribution itself also encapsulates uncertainty information (e.g., variance and confidence intervals). For instance, if one agent’s belief is highly concentrated (due to sufficient observational data) while another’s is relatively dispersed (due to insufficient data), users will receive inconsistent answers when repeatedly querying different agents about the same knowledge. This degrades the user experience, and solving this problem requires direct exchange of beliefs among agents.

Achieving group-consistent cognitive KBs brings two key benefits to EQA:

- More reliable task decisions. Consistent cognitive KBs ensure that all agents perform reasoning based on the same understanding of the scene, avoiding conflicts or errors caused by divergent individual beliefs, thus generating more reliable answers.

- More efficient group communication. Once the individual cognitive KBs in the group reach consistency, the KB can be organized in a way such that the same knowledge indices within them are shared by all agents. During communication, agents need only transmit these indices instead of raw natural language or semantics, drastically reducing communication overhead.

B. Cognitive Coordination Framework

To achieve accurate estimation and global consensus, we propose a Distributed Bayesian Cognitive Knowledge Coordination (**DisBayesCKC**) framework. In this framework, each agent is treated as a distributed sensing node that observes the environment from its local perspective and forms local posterior beliefs about a subset of entity relationships. Through periodic exchange and fusion of beliefs, the system effectively integrates multi-source local observations, and builds accurate and globally consistent group cognitive KBs progressively. The workflow consists of:

- Local observation and Bayesian update. At each time step, agents detect entity relationships within their perceptual range, obtaining binary observations. They update the posterior distribution for corresponding relationships in their cognitive KB using Bayes' rule, refining the accuracy of individual belief.
- Neighbor communication and belief fusion. Agents send updated posterior distribution parameters to communication neighbors. Upon receiving neighbors' beliefs, agents employ a fusion algorithm to integrate external knowledge with their own, reducing local observation bias and increasing estimation confidence. Iterating this "observe-update-communicate-fuse" cycle drives all agents' cognitive KBs toward consistency, ultimately converging to a unified description of the environment's objective true state.

1) *Local Observation and Bayesian Update*: At time step t , suppose agent a_i observes relationship u_k and obtains observation $x_{i,k}^t \in \{0,1\}$, where 1 indicates the relationship is observed to hold and 0 otherwise. Assuming independent observations, $x_{i,k}^t$ follows a Bernoulli distribution: $Bernoulli(\theta_k^*)$. Each agent maintains a Beta distribution as the posterior belief: $p(\theta_{i,k}^t | x_{i,k}^t) \sim Beta(\alpha_{i,k}^t, \beta_{i,k}^t)$. The Beta distribution $Beta(\alpha, \beta)$, defined on $[0, 1]$, naturally models the probability of binary events like relationship existence, with $\alpha - 1$ and $\beta - 1$ commonly interpreted as success and failure counts. Furthermore, in Bayesian estimation, the Beta distribution serves as the conjugate prior for the Bernoulli likelihood. This conjugacy facilitates a recursive learning process where the posterior belief from the previous time step $t-1$ functions directly as the prior for time step t . Specifically, let the agent's prior belief be modeled as $p(\theta_{i,k}^{t-1}) \propto \theta_{i,k}^{\alpha_{i,k}^{t-1}-1} (1-\theta)^{\beta_{i,k}^{t-1}-1}$. Upon acquiring a new binary observation $x_{i,k}^t$ with likelihood $L(\theta) \propto \theta^{x_{i,k}^t} (1-\theta)^{1-x_{i,k}^t}$, the current posterior is derived by multiplying the prior and the likelihood:

$$p(\theta_{i,k}^t | x_{i,k}^t) \propto \theta^{(\alpha_{i,k}^{t-1} + x_{i,k}^t) - 1} (1-\theta)^{(\beta_{i,k}^{t-1} + 1 - x_{i,k}^t) - 1}. \quad (3)$$

Algorithm 1 Multi-Agent DisBayesCKC.

Input: Agent set $\mathcal{A} = \{a_1, a_2, \dots, a_N\}$, triple set $\mathcal{U} = \{u_1, u_2, \dots, u_M\}$, topology graph $G = (\mathcal{V}, \mathcal{E})$
Output: Estimated distribution $Beta(\alpha_{i,k}, \beta_{i,k})$ of each agent for each relationship

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1: Initialize for all  $a_i \in \mathcal{A}, u_k \in \mathcal{U} : \alpha_{i,k} \leftarrow 1, \beta_{i,k} \leftarrow 1$ 
2: Initialize for all  $a_i \in \mathcal{A} : is\_Initiator[i] \leftarrow false$ 
3:  $t = 1, T = T_0$ 
4: while true do
5:   if agent positions change then
6:     Update communication topology  $G^t = (\mathcal{A}, \mathcal{E}^t)$ 
7:     Recompute Perron matrix  $P^t$ 
8:     For each  $a_i \in \mathcal{A}$ : update its neighbor set  $\mathcal{N}(i)$ 
9:   end if
10:   $a_i \leftarrow \text{random\_choice}(\mathcal{A})$  //observation phase
11:  for each  $u_k$  in  $\mathcal{U}$  do
12:     $x_{i,k}^t \leftarrow \text{observe\_event}(a_i, u_k)$ 
13:     $\alpha_{i,k}^t \leftarrow \alpha_{i,k}^{t-1} + x_{i,k}^t, \beta_{i,k}^t \leftarrow \beta_{i,k}^{t-1} + 1 - x_{i,k}^t$ 
14:     $is\_Initiator[i] \leftarrow true$ 
15:  end for
16:  if  $t \% T = 0$  then // communication phase
17:     $initiators \leftarrow \{a_i \in \mathcal{A} | is\_Initiator[i] = true\}$ 
18:    for each  $a_i$  in  $initiators$  do in parallel
19:      For each  $a_j \in \mathcal{N}(i)$ :  $\text{send\_message}(j, \{(\alpha_{i,k}^t, \beta_{i,k}^t) \text{ for } u_k \in \mathcal{U}\})$ 
20:    end for
21:    for each  $a_i$  in  $\mathcal{A}$  do in parallel // fusion phase
22:       $received\_messages \leftarrow \text{wait\_messages}(\mathcal{N}(i), \text{timeout})$ 
23:      For each  $u_k \in \mathcal{U}$ :  $(\alpha_{i,k}^t, \beta_{i,k}^t) \leftarrow \text{LogP}(received\_messages, P^t)$ 
24:    end for
25:    Reset  $is\_Initiator[]$ 
26:  end if
27:   $t \leftarrow t + 1$ 
28: end while
```

The resulting form is clearly recognized as the kernel of a new Beta distribution, confirming that the posterior preserves the Beta structure. Consequently, the parameters are updated simply by adding the observation counts:

$$\alpha_{i,k}^t = \alpha_{i,k}^{t-1} + x_{i,k}^t, \quad \beta_{i,k}^t = \beta_{i,k}^{t-1} + 1 - x_{i,k}^t. \quad (4)$$

As observations accumulate, α^t and β^t increase, sharpening the distribution to reflect higher confidence. Initially, each agent's prior is set to an uninformative $Beta(1, 1)$.

2) *Neighbor Communication and Belief Fusion*: Let the communication topology among agents be represented by a graph $G^t = (\mathcal{A}, \mathcal{E}^t)$, where edge set $\mathcal{E}^t = \{l_1, l_2, \dots, l_L\}$ represents communication links. Agent mobility causes link changes over time and we assume G^t remains connected. We define a fixed-length time interval as an observation period. During this period, agents independently perform observations and posterior updates. At the end of the period, a communication-and-fusion phase is triggered: agents that performed observations during the period initiate communication

by sending their posterior parameters to their neighbors. Upon receiving these parameters, each agent fuses the incoming beliefs with its own and updates its estimate accordingly. For posterior parameter fusion, we use the Logarithmic Opinion Pool (LogOP) method [34]. LogOP constructs the aggregated distribution as the geometric weighted average of individual probability densities:

$$p_{\text{agg}}(\theta) \propto \prod_{j=1}^n p_j(\theta)^{w_j}. \quad (5)$$

Let $\mathcal{N}(i)$ denote the neighbor set of agent a_i . During fusion, agent a_i receives Beta posteriors from its neighbors. For the set of m Beta densities (including its own), denoted by $\{p_l(\theta_{l,k}^t)\}_{l \in \mathcal{N}(i) \cup \{a_i\}}$, where each density is given by $p_l(\theta_{l,k}^t) \propto \theta^{\alpha_{l,k}^t - 1} (1 - \theta)^{\beta_{l,k}^t - 1}$, agent a_i computes fused distribution parameters. Substituting the Beta probability density functions into the LogOP formulation Eq. (5), the aggregated density is derived as the product of weighted powers:

$$\begin{aligned} p_{\text{agg}}(\theta) &\propto \prod_j [\theta^{\alpha_j - 1} (1 - \theta)^{\beta_j - 1}]^{w_j} \\ &= \theta^{\sum w_j (\alpha_j - 1)} (1 - \theta)^{\sum w_j (\beta_j - 1)}. \end{aligned} \quad (6)$$

This aggregated form clearly corresponds to the kernel of a new Beta distribution $\text{Beta}(\sum w_j \alpha_j, \sum w_j \beta_j)$ (note that $\sum w_j = 1$). Thus, the fusion yields updated parameters calculated as the weighted sum:

$$\begin{aligned} \alpha_{\text{agg}}^t &= \sum_{a_j \in \mathcal{N}(i) \cup \{a_i\}} w_j \alpha_{j,k}^t, \\ \beta_{\text{agg}}^t &= \sum_{a_j \in \mathcal{N}(i) \cup \{a_i\}} w_j \beta_{j,k}^t. \end{aligned} \quad (7)$$

For weight selection, we adopt the method from [35], using the Perron matrix $P = (I - \epsilon D) + \epsilon A$ as the weight table for N agents. $P[i, j], a_i, a_j \in \mathcal{A}$ denotes the weight assigned to agent a_j 's parameters when fused by agent a_i , where I is the identity matrix, $A \in \mathbb{R}^{N \times N}$ is the adjacency matrix of G^t ($A[i, j] = 1$ if link exists, else 0), $D = \text{diag}\{d_1, d_2, \dots, d_N\}$ is the degree matrix (d_i is the neighbor count of a_i), and $\epsilon \geq 0$ is the discrete step size. The LogOP-based fusion mechanism enables local belief alignment through weighted averaging, while the connectivity of the communication topology ensures that local consensus propagates throughout the entire agent network, eventually achieving global consistency. LogOP creates a collaborative learning loop: since the fused distribution remains Beta, it directly serves as the prior for the next observation cycle. The form of the posterior distribution remains consistent throughout the iterative process, enabling seamless integration of individual-level Bayesian learning and group-level consensus formation.

The DisBayesCKC process is outlined in Algorithm 1. Lines 1-2 initialize each estimate as a Beta(1,1) distribution, and the *is_Initiator*[] array flags which agents act as communication initiators during the collaborative process. Lines 10-15 perform observation and update using Eq. (4). In lines 18-20, agents initiate communication in parallel. Lines 21-24 describe

that after a waiting period of duration *timeout* for message reception, each agent performs fusion updates. The LogP(.) function in line 23 takes the caller's own posterior and received parameters as inputs, queries the weight matrix P^t , and applies Eq. (7) to compute the fused distribution.

C. Scalability in Large Knowledge Spaces

The triplet set U in this paper is a candidate knowledge space, which is used to define belief representation, Bayesian update, and group consensus objectives. It does not require each agent to explicitly maintain the complete U in practical operation. When $|U|$ is large, assigning Beta parameters to all triplets and transmitting the complete posterior state in every communication round will cause high storage, computation, and communication overhead.

In practical implementation, DisBayesCKC can use sparse maintenance. For agent a_i , let O_i^t denote the set of triplets observed from its local perspective, and let C_i^t denote the set of triplets received from its neighbors. Then, the active set explicitly maintained by the agent is defined as $U_i^t = O_i^t \cup C_i^t$, $U_i^t \subseteq U$. Triplets in U_i^t are assigned Beta parameters and are involved in Bayesian update or LogOP fusion. For triplets $u_k \notin U_i^t$, their beliefs are not explicitly instantiated and are kept as the uninformative prior Beta(1, 1). In addition, the update and communication processes of DisBayesCKC are triggered on demand in practical systems. The agent only applies the Bayesian update in Eq. (4) to the triplets actually observed in the current observation period. During communication, it only sends the posterior parameters of these newly observed triplets, instead of sending the complete belief state over U . After receiving the corresponding parameters, the neighbor agent inserts them into its own active set and performs fusion according to Eq. (7). In this way, the system does not take the theoretical candidate space U as the direct object of computation and communication in every round. Instead, maintenance, update, and transmission are restricted to the knowledge subset activated by local observations and neighbor communication. Since the core mechanism of DisBayesCKC is based on local Bayesian update and distributed belief fusion, it does not require any single agent to fully observe the global knowledge. Therefore, even when the KB is large, observations are sparse, and no omniscient agent exists, the basic inference process still holds. This design is also more consistent with practical large-scale KB scenarios, where knowledge is sparsely distributed and observations occur locally. The sparse observation experiment in VII-B further verifies the effectiveness of DisBayesCKC under this implementation.

V. HIGH-LEVEL SEMANTIC EXTRACTION

A. Triplets Decomposition Based on the Cognitive KB

EQA tasks are highly scene-dependent, meaning Q&A require grounding entities, attributes, and relations in the physical scene. Based on semantic intent, EQA task sentences can be categorized into several illustrative types, as shown in Table I. The common semantic feature across EQA task sentences is that both questions and answers consistently refer to

TABLE I
CATEGORIZATION OF SENTENCES IN EQA TASKS

Category	Description	Examples	High-level semantics
Attribute	Inquiring about or describing attributes of an object (color, material, etc.)	“What color is the mug?”/ “The coffee brand on the office table is Luckin.”	(mug, color, ?) / (table, in, office); (coffee, on, table); (coffee, brand, Luckin)
Spatial-Relational	Inquiring about or describing the location or relative relationship of an object	“Where is that blue stapler?”/ “The calculator is on the table in the boardroom.”	(stapler, color, blue); (stapler, in, ?) / (table, in, boardroom); (calculator, on, table)
Boolean (Yes/No)	Determining or confirming whether a certain state or condition holds	“Is the light on the dining table next to the kitchen turned off?”/ “The light in the bedroom is on.”	(dining table, in, kitchen); (light, on, dining table); (light, ?, turn off) / (light, in, bedroom); (light, state, on)
Counting	Inquiring about or stating the quantity of objects	“How many chairs are around the table?”/ “There are two envelopes on the table in the lobby.”	(chair, around, table); (chairs, quantity, ?) / (table, in, lobby); (envelope, on, table); (envelopes, quantity, two)

specific entities, relationships, and attributes within the scene. Therefore, sentence parsing and reasoning require grounding language symbols to corresponding nodes or edges in the scene representation (e.g., a cognitive KB). This grounding enables the EQA system to transition from language understanding to semantic reasoning and perceptual decision-making.

In semantic communication, the cognitive KB provides agents with a unified semantic space. Decomposing natural language into triplets, which form the cognitive KB’s fundamental units, enables the system to explicitly identify the semantic units and their interrelationships. Based on whether triples exist in the cognitive KB, the system determines which semantics are already shared and which need to be fully transmitted as new semantic information.

In EQA tasks, the extracted triplets can be expressed as

$$\mathcal{U} = \{\mathcal{U}^{\text{known}}, \mathcal{U}^{\text{unknown}}\}$$

where triplets in the triplets set $\mathcal{U}^{\text{known}}$ already exist in the cognitive KB, whereas those in the triplets set $\mathcal{U}^{\text{unknown}}$ are absent from it. Triplets in $\mathcal{U}^{\text{known}}$ are used to locate queried or referenced entities within the cognitive KB. The semantics contained in these triplets are already explicitly present in the cognitive KB. For such triplets, only their indices need to be transmitted and the receiver then retrieves the full triplets locally. In contrast, triplets in $\mathcal{U}^{\text{unknown}}$ must be transmitted in full, guaranteeing the complete semantic recovery of information at the receiver.

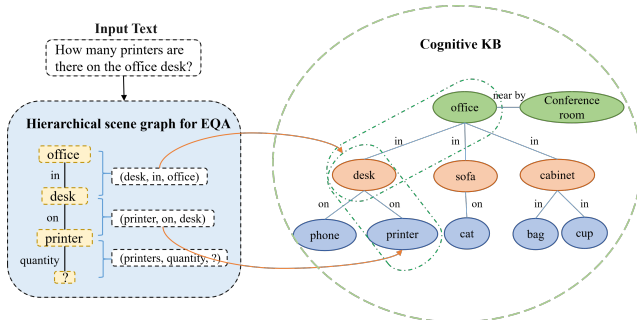


Fig. 4. Semantic triplets decomposition based on the cognitive KB.

As shown in Fig. 4, the triplets $(\text{desk}, \text{in}, \text{office})$, $(\text{printer}, \text{on}, \text{desk}) \in \mathcal{U}^{\text{known}}$. Therefore, during agent commu-

nication, only the indices of these existing triplets and the full content of the triplet $(\text{printers}, \text{quantity}, ?) \in \mathcal{U}^{\text{unknown}}$ need to be transmitted. This enables semantic completeness while significantly lowering communication overhead.

B. EQA Multimodal Semantic Extraction and Transmission Framework

Building on the hierarchical structure of the cognitive KB, this paper introduces an EQA multimodal semantic extraction and transmission framework supporting bidirectional text-triplet conversion and unidirectional triplet extraction from graphical inputs.

- **Text-triplet bidirectional conversion:** Modeled as a sequence-to-sequence task and handled by T5-based models.
- **Graphic-triplet extraction:** Processed under the “Relation Conversation” paradigm of the All-Seeing Model V2 [36], which produces visually grounded structured text that can be directly parsed for triplet extraction.

The overall processing framework for EQA tasks is illustrated in Fig. 5.

Specifically, in the text-to-triplet semantic extraction process, the T5 Text2Triplet model receives input text and outputs semantic triplets. Together with the cognitive KB, it determines which triplets are transmitted as indices and which require full transmission. As illustrated in Fig. 5, given the question “How many blue pillows are on the sofa in the living room?”, the semantic triple sequence “Index 1; Index 2; (pillows, quantity, ?)” is generated, where Index 1 refers to (sofa, in, living room) and Index 2 to (pillow, color, blue). Similarly, in the triplet-to-text subtask, the T5 Triplet2Text model, integrated with the cognitive KB, converts the triplets sequence into natural language.

For the triplet extraction from images, the All-Seeing Model V2 takes both textual prompting and the perceived scene image as inputs. It respectively extracts visual and textual features through a visual encoder and a language model, aligns the two modalities via a vision–language connector, and generates a structured response sequence in an autoregressive manner [36]. The generated sequence is then integrated with the cognitive KB to produce the extracted semantic triplets sequence to be

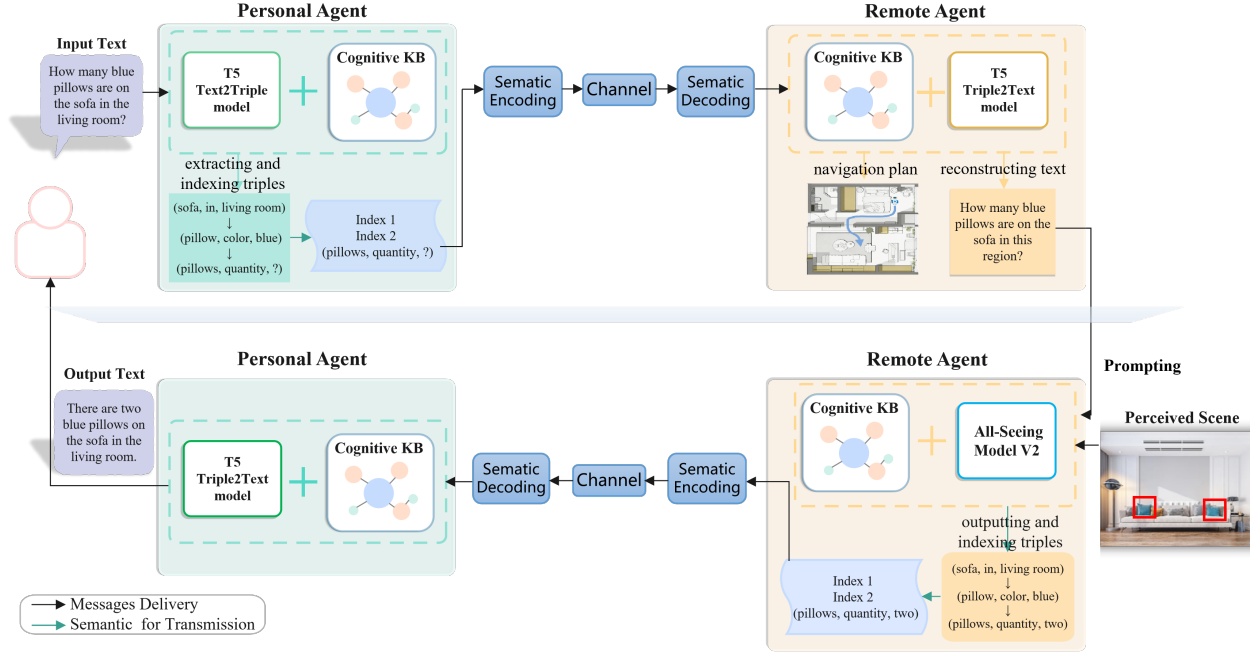


Fig. 5. The multimodal semantic extraction and transmission framework in collaborative EQA.

transmitted. As shown in Fig. 5, the output sequence “Index 1; Index 2; (pillows, quantity, two)” is generated.

C. Agent Implementation

For real agent implementation consideration. The T5-small model has 60M parameters (230 MB GPU memory), while the All-Seeing Model V2 has 13B parameters (27 GB GPU memory). As IoT terminals, robots differ from general IoT devices in computation and power. For EQA tasks, the VLM is mainly used for question answering rather than navigation. On an A6000 GPU, its average QA latency is 1.42 seconds per image with power consumption of 250–300 W, which is acceptable in this scenario. Lighter VLM models and embedded AI platforms, such as the NVIDIA Jetson AGX Orin 64GB (275 TOPS, 64GB memory), can also be considered. Moreover, models like SpatialBot-3B [37] demonstrate strong scene understanding. Given current trends, deploying powerful yet lightweight small models on robot terminals will become increasingly feasible.

VI. TRIPLETS SEMANTIC SOURCE AND CHANNEL ENCODING

The above sections systematically elaborate on the high-level semantic extraction and triplet transmission mechanisms based on cognitive KB for the collaborative EQA tasks. By mapping natural language sentences and multimodal perception into semantic triplets and implementing differential transmission based on their known status in the Cognitive KB, the system can significantly reduce communication redundancy while ensuring semantic consistency.

However, the aforementioned mechanisms primarily focus on “which semantic units to transmit”, without yet addressing

how these semantic units are efficiently and robustly represented and transmitted over physical channels. To this end, this section details the semantic encoding and transmission mechanism, tailored for transmitting triplets, with entity embedding, compression, and quantization of continuous semantic representations, as well as their joint optimization over noisy channels, thereby realizing a complete closed loop from high-level semantic decision-making to low-level physical transmission. We denote the proposed mechanism as **Triplet-EmPress**, which performs triplets embedding and compression for agentic semantic communication.

A. Overall System Architecture

Fig. 6 illustrates the overall architecture of the proposed semantic communication system with four modules: entity embedding, transmitter (semantic and channel encoder), receiver (semantic and channel decoder), entity reconstruction and semantic utility.

Entity Embedding. We define an word space with all the evolved entities in the EQA task. The entity e_i is first mapped to a continuous semantic space via the Word2Vec model to obtain the initial semantic representation T_i . This module maps discrete symbolic entities into continuous vectors suitable for neural network processing, serving as the input for subsequent semantic compression and transmission.

Transmitter. The transmitter is composed of Fully Connected Network (FCN) and Highway Gate, utilized to perform nonlinear feature transformation and compression on the input semantic vectors to obtain the compressed semantic representation H . Subsequently, the compressed semantic vectors undergo quantization mapping via a threshold function. This module implements feature extraction, dimensionality reduction, and vector quantization at the semantic level, providing

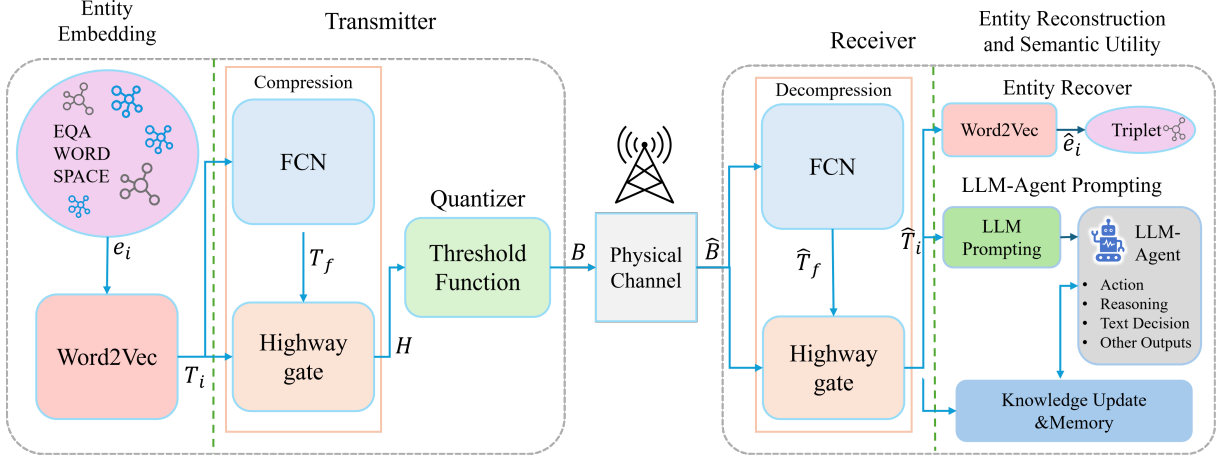


Fig. 6. Semantic Source and Channel Encoding for agentic SC.

the foundation for subsequent low-bandwidth transmission. Typically we consider the Joint Source Channel Coding (JSCC) scheme in which the noise and fading of physical channel are modeled.

Receiver. The receiver structure is symmetric to the transmitter, consisting of FCN and Highway Gate. It performs dequantization and semantic reconstruction on the received bitstream to recover the continuous semantic representation $\hat{T}_i$. The training objective of this module is to minimize the reconstruction error between the original semantic representation T_i and the reconstructed semantic representation $\hat{T}_i$ to ensure consistency in the semantic space.

Entity Reconstruction and Semantic Utility. This module takes the reconstructed semantic representation $\hat{T}_i$ as input and supports various downstream semantic utilization, including:

- **Entity Recovery.** By utilizing Word2Vec inverse mapping or nearest neighbor search, $\hat{T}_i$ is mapped back to the entity space to obtain the reconstructed entity $\hat{e}_i$, which is used for triplet recovery and knowledge graph reasoning tasks.
- **LLM-Agent Prompting.** $\hat{T}_i$ is directly input into the LLM agent as a structured semantic state to realize high-level semantic tasks such as reasoning, decision-making, and text generation.
- **Knowledge Update & Memory.** $\hat{T}_i$ is written into external memory or knowledge bases for long-term knowledge updates, personalized modeling, and subsequent semantic retrieval.

Through the aforementioned design, the system not only supports traditional entity-level semantic recovery tasks but also enables task-oriented agentic semantic communication oriented towards agent collaboration and diverse semantic application scenarios.

B. Entity Embedding

This work adopts the Skip-gram model under the Word2Vec framework to model the embedding of knowledge graph entities, primarily based on its low implementation complexity, strong decoupling in training, and good engineering scalability.

In triplets transmission, the $\mathcal{U}^{\text{known}}$ set is transmitted by their indices, hence the main focus of entity embedding is on the out-of-KB triples, the $\mathcal{U}^{\text{unknown}}$ set. The cognitive KB is a structure KG, however entities in the $\mathcal{U}^{\text{unknown}}$ set are often one-time, non-declarative, contain special symbols, or take the form of questions. Such content often does not constitute a statement or fact suitable for inclusion in a structured KB, but word2vec can be used to capture its semantics in a lighter and more general way. As a mature and lightweight embedding method, Word2Vec can efficiently generate continuous semantic representations without introducing complex network structures, thereby reducing the overall system implementation cost. Furthermore, this method supports a “two-stage independent training” paradigm, where entity embeddings can be pre-trained offline, and the semantic compression and transmission modules can be optimized independently on a fixed embedding space, thus avoiding the instability and computational overhead associated with joint training. In addition, the semantic vectors generated by Word2Vec possess good interpretability, and the inference phase involves only vector lookups and similarity calculations, resulting in low resource demands and high computational efficiency, making it suitable as the semantic source representation method for the semantic communication system.

Serving as the semantic source signal, entity embedding acts as the continuous semantic representation within the semantic space and constitutes the core input of the semantic communication system. The to be transmitted triplet is represented as $u_k = (e_{1k}, r_k, e_{2k})$. Here, e_{1k} and e_{2k} represent the head entity and the tail entity, respectively, and r_k represents the relationship between the entities. The semantic space of the working scenario is denoted as $\mathcal{U} = \{u_k\}_{k=1}^N$, where N is the total number of triplets. Consequently, the entity set can be defined as

$$\mathcal{E} = \{e \mid e = e_{1k} \text{ or } e = e_{2k}, \forall u_k \in \mathcal{U}\}. \quad (8)$$

The Skip-gram model of Word2Vec is utilized to train the entity embeddings. For a center entity e_c and a context entity

e_o , the conditional probability is defined as

$$p(e_o | e_c; \theta) = \frac{\exp(\mathbf{v}_{e_o}^\top \mathbf{v}_{e_c})}{\sum_{e \in \mathcal{E}} \exp(\mathbf{v}_e^\top \mathbf{v}_{e_c})}. \quad (9)$$

where $\mathbf{v}_e \in \mathbb{R}^D$ denotes the embedding vector of entity $e \in \mathcal{E}$ ($D = 300$), and $\mathbf{v}_{e_c}$ and $\mathbf{v}_{e_o}$ are the embeddings of the center entity e_c and the context entity e_o , respectively.

Skip-gram performs training by maximizing the conditional likelihood of entity pairs in the knowledge graph, with the optimization objective given by

$$\max_{\theta} \sum_{u_k \in \mathcal{U}} \log p(e_{2k} | e_{1k}; \theta). \quad (10)$$

Upon completion of training, each entity is mapped to a dense vector space, and its embedding vector undergoes L_2 normalization:

$$T_i = \frac{\mathbf{v}_e}{\|\mathbf{v}_e\|_2}, \quad e \in \mathcal{E}. \quad (11)$$

The normalized vector $T_i \in \mathbb{R}^D$ serves as the input for the semantic communication system. Let the reconstructed continuous semantic vector be denoted as $\hat{T}_i$ and the reconstructed entity as $\hat{e}_i$. By maintaining an entity embedding vocabulary at the receiver that is consistent with that of the transmitter, the process of mapping $\hat{T}_i$ back to the discrete entity space can be abstractly represented as:

$$\hat{e}_i = \text{query}(\hat{T}_i), \quad (12)$$

where $\text{query}(\cdot)$ represents the retrieval function within the word space, and its output $\hat{e}_i$ is the entity closest to $\hat{T}_i$.

C. Transmitter

The objective of the transmitter module is to compress high-dimensional continuous semantic vectors into low-dimensional discrete representations, while maximizing the preservation of their core semantic information. This module consists of two cascaded sub-modules: the *Semantic Compression Module* and the *Semantic Quantization Module*.

1) *Semantic Compression*: The input $T_i \in \mathbb{R}^D$ first undergoes a nonlinear transformation through a Fully Connected Network (FCN):

$$T_f = \text{ReLU}(T_i W_f + b_f) \quad (13)$$

where the weight matrix of this layer is $W_f \in \mathbb{R}^{D \times K}$, the bias vector is $b_f \in \mathbb{R}^K$, and the output $T_f \in \mathbb{R}^K$ is the preliminarily extracted semantic feature representation. D represents the input dimension, and K represents the compression dimension.

Subsequently, adaptive information fusion is performed through a Highway Network. The Highway Network allows the model to traverse multi-hop linear paths within the deep structure, thereby enhancing its expressive capability. The transform gate G is calculated by the Sigmoid function:

$$G = \sigma(T_i W_g + b_g) \quad (14)$$

where $W_g \in \mathbb{R}^{D \times K}$ and $b_g \in \mathbb{R}^K$ are the learnable parameters of the transform gate. The output of the Highway Network is:

$$H = G \odot T_f + (1 - G) \odot T_i W_a \quad (15)$$

where $W_a \in \mathbb{R}^{D \times K}$ is the learnable parameter used for dimension alignment. The Sigmoid function restricts the output to the interval $(0, 1)$, allowing G to serve as a "gating signal". This gating mechanism enables the network to adaptively choose between retaining original information or utilizing transformed features across different dimensions, thereby effectively alleviating the vanishing gradient problem.

2) *Semantic Quantization*: The objective of the semantic quantization module is to convert the continuous floating-point vector H into a discrete binary vector B , thereby achieving extreme information compression. Consequently, each floating-point number (32-bit) is compressed into 1 bit. The quantization module converts the continuous vector $H \in \mathbb{R}^K$ into the binary vector $B = [b_1, b_2, \dots, b_K]^\top \in \{0, A\}^K$, realizing compression (with a compression ratio of 32:1). The quantization function is defined as:

$$b_i = \begin{cases} A, & \text{if } H_i \geq q_{th} \\ 0, & \text{otherwise} \end{cases} \quad (16)$$

where H_i represents the i -th dimension of H , and the quantization function threshold q_{th} is designed as the average value of H_i across the K dimensions.

$$q_{th} = \frac{1}{K} \sum_{i=1}^K H_i \quad (17)$$

D. Physical Channel

For the quantized discrete vector B , this paper adopts a unified multiplicative fading and additive noise model to characterize the wireless physical channel. The received signal can be expressed as

$$Y = hB + \mathbf{n} \quad (18)$$

where h denotes the channel gain coefficient, and $\mathbf{n}$ represents the Additive White Gaussian Noise (AWGN) vector, satisfying $\mathbf{n} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$.

When h equals 1, the model reduces to a standard AWGN channel; when h follows a Rayleigh distribution, it corresponds to a Rayleigh fading channel. By adjusting the noise variance σ^2 , the transmission performance under different Signal-to-Noise Ratio (SNR) conditions can be simulated. The recovered binary semantic vector $\hat{B} = [\hat{b}_1, \hat{b}_2, \dots, \hat{b}_K]^\top \in \{0, A\}^K$ is obtained by performing equalization and decision operations on Y .

E. Receiver

The objective of the receiver is to accurately recover the original continuous semantic vector $\hat{T}$ from the discrete vector $\hat{B}$, which has undergone quantization and channel perturbation and may contain errors. The receiver adopts an architecture **symmetric** to that of the transmitter, yet its parameters are independent of the transmitter. It consists of a fully connected layer and a Highway Network. The received vector $\hat{B}$ is first processed by the fully connected layer and then mapped back to the original dimension through the Highway Network:

$$\hat{T}_i = \text{decoder}(\hat{B}) \quad (19)$$

The output of the fully connected layer is:

$$\hat{T}_f = \text{ReLU}(\hat{B}\hat{W}_f + \hat{b}_f), \quad (20)$$

where $\hat{W}_f \in \mathbb{R}^{K \times D}$, $\hat{b}_f \in \mathbb{R}^D$, and D is the dimension of the original semantic vector.

The output of the Highway Network is:

$$\hat{G} = \sigma(\hat{B}\hat{W}_g + \hat{b}_g) \quad (21)$$

$$\hat{T}_i = \hat{G} \odot \hat{T}_f + (1 - \hat{G}) \odot \hat{B}\hat{W}_a, \quad (22)$$

where $\hat{W}_a \in \mathbb{R}^{K \times D}$, $\hat{G}$ is the gating signal, and $\hat{W}_g$ and $\hat{b}_g$ are learnable parameters.

The output of the Highway Network serves as the final estimated semantic vector $\hat{T}_i \in \mathbb{R}^D$ of the decoder, which is also the reconstruction estimate of the original semantic vector T_i .

It is worth noting that the step quantization function adopted in the previous subsections belongs to discontinuous mappings, and its derivative is zero almost everywhere in the domain, which would cause the gradient update mechanism based on backpropagation to fail. To overcome this issue, this paper introduces the Straight-Through Estimator (STE) mechanism, which recovers gradient information for backpropagation through approximation methods while ensuring that the real quantization operation is performed during forward propagation.

The continuous semantic representation before quantization is H , and the received quantized vector obtained after channel transmission and noise perturbation is $\hat{B}$, which is a distorted version of the original binary vector B . In the forward propagation phase, the decoder uses $\hat{B}$ as input for semantic reconstruction; while in the backward propagation phase, STE is utilized to approximate the non-differentiable quantization and channel perturbation processes as an identity mapping, allowing the gradient to bypass $\hat{B}$ and act directly on the continuous variable H .

The system takes the Mean Squared Error (MSE) between the reconstructed semantic vector and the original semantic vector as the training objective, and its loss function is defined as

$$L = \text{MSE}\left(T_i, \text{decoder}\left(H + \text{sg}\left[\hat{B} - H\right]\right)\right), \quad (23)$$

where T_i represents the original continuous semantic representation, $\text{decoder}(\cdot)$ is the semantic decoding network, and $\text{sg}[\cdot]$ denotes the stop-gradient operator.

During the forward propagation process,

$$H + (\hat{B} - H) = \hat{B},$$

thus the network actually uses the distorted semantic bits output by the channel for reconstruction; while in the backward propagation phase, since $\text{sg}[\cdot]$ blocks the influence of $\hat{B}$ on the gradient, the gradient propagates only through the continuous variable H , thereby avoiding the training difficulties caused by the non-differentiability of the quantization and channel models.

VII. EVALUATION

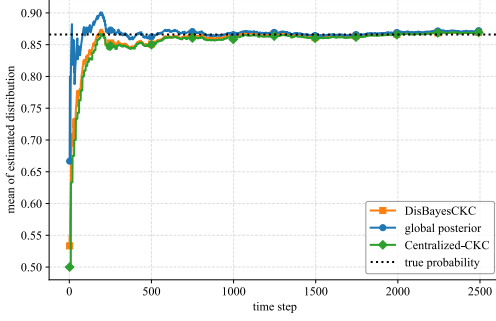
We evaluate the mechanism proposed in this paper from the following three aspects. First, we evaluate the performance of the proposed DisBayesCKC mechanism on achieving cognitive consistency in multi-agent groups. Second, we elaborate on the training data preparation and model training performance for the text-to-triplet model. Finally, we validate the transmission effectiveness of the proposed Triplets-EmPress joint semantic source-channel encoding mechanism. The following evaluations are conducted on the same EQA dataset, which consists of 1987 EQA question-answer samples.

A. Cognitive KB Coordination Performance

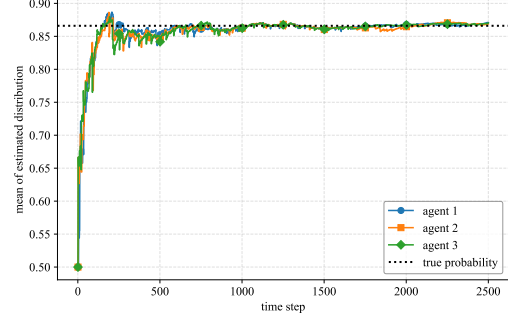
We consider an EQA system comprising $N = 10$ agents, where all agents are jointly concerned with $M = 1500$ relationships. Each relationship is associated with an underlying true probability of validity. In the simulated data, these probability values are distributed as follows: 10% in the interval $(0, 0.05]$, 10% in $[0.95, 1)$, and 80% in $(0.05, 0.95)$. This distribution reflects that a small portion of relationships are nearly fixed (e.g., “salt is on the coffee table” with low probability or “pillow is on the bed” with high probability), while most relationships are influenced by daily user activities (e.g., “thermos cup is in the kitchen/coffee table/room”). Since users are more likely to inquire about uncertain relationships, such cases dominate the simulated dataset. Let τ denote a time step, with a total simulation duration of 2500τ . At each time step, one randomly selected agent observes all triplets. The observation result is generated by uniform sampling in $[0, 1]$: if the sample value exceeds the true probability of the triplet, the observation result is 1 (valid); otherwise, it is 0 (invalid). This ensures the observations follow a Bernoulli distribution. The communication interval is 5τ . During communication, all transmitted posterior parameters propagate 2 hops within the topology.

The communication topology among agents significantly affects the collaborative efficiency. To evaluate the generality of our proposed algorithm, we consider four typical network topologies: Barabasi-Albert scale-free network [38], Watts-Strogatz small-world graph [39], star graph, and ring graph. To comprehensively assess the performance of our DisBayesCKC framework, we establish four baseline methods for comparison:

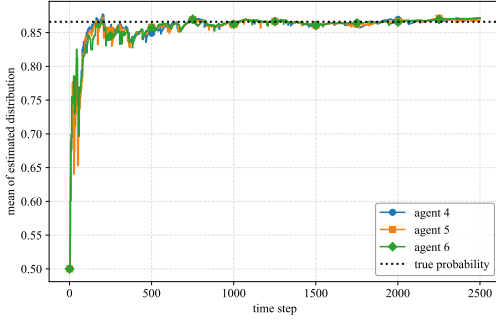
- 1) *World Observer (W-Obs)*. Assumes an omniscient world observer that has access to all observations from all agents across all time steps. Using this full dataset, it updates a non-informative Beta(1,1) prior to obtain a posterior distribution. This represents the upper bound of accuracy for using Bayesian updates but is infeasible in practical distributed systems.
- 2) *Centralized Cognitive Knowledge Coordination (Cent-CKC)*. Assumes a central server. At each communication round, all agents transmit their local posterior parameters to the server, which fuses them using the LogOP method described in Section IV-B2. The fused global posterior is then broadcast back to all agents. This approach sacrifices communication efficiency to achieve near-optimal



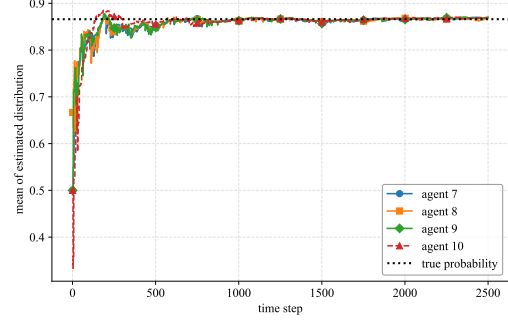
(a) Performance comparison among DisBayesCKC, global posterior, and Centralized-CKC.



(b) Estimation trajectories of Agents 1, 2, and 3.



(c) Estimation trajectories of Agents 4, 5, and 6.



(d) Estimation trajectories of Agents 7, 8, 9, and 10.

Fig. 7. Temporal evolution of the estimated mean distribution, comparing the proposed DisBayesCKC method with baselines and illustrating individual agent convergence.

estimation and serves as a performance reference for distributed methods.

- 3) *Independent Observers (Ind-Obs)*. Agents update their beliefs based solely on their own local observations, without participating in communication or belief fusion with other agents. This baseline is used to reflect the information gain and performance improvement brought by collaborative communication.
- 4) *Beta-Dempster-Shafer with Feedback (BetaDS-FB)*. This baseline is adapted from the networked information fusion method proposed by Liang et al. [40]. While the original method focuses on decision-making, we extend it to a hybrid architecture for probability estimation. In this framework, agents maintain local beliefs as Beta distributions but map them to Basic Belief Assignments during communication. Adopting the core mechanism, a credibility test based on Jousselme distance is employed to construct an interaction graph that filters out highly conflicting neighbor evidence before fusion. The validated evidences are then aggregated using the Mean Combination Rule to avoid the “consensus amplification” of Dempster’s rule.

The estimation accuracy is evaluated using the Mean Absolute Error (MAE) between either the mean or the mode of the estimated distribution and the true probability. For agent a_i and triplet u_k , let θ_k^* denote the true validity probability, and let $\hat{\theta}_{i,k}$ denote the point estimate obtained from the posterior

distribution $p(\theta_{i,k})$. The MAE is defined as

$$\text{MAE} = \frac{1}{NM} \sum_{i=1}^N \sum_{k=1}^M |\hat{\theta}_{i,k} - \theta_k^*|. \quad (24)$$

The point estimate $\hat{\theta}_{i,k}$ can be instantiated as either the posterior mean or the posterior mode. For a posterior distribution $\text{Beta}(\alpha_{i,k}, \beta_{i,k})$, the posterior mean is $\hat{\theta}_{i,k}^{\text{mean}} = \frac{\alpha_{i,k}}{\alpha_{i,k} + \beta_{i,k}}$, and the posterior mode is $\hat{\theta}_{i,k}^{\text{mode}} = \frac{\alpha_{i,k} - 1}{\alpha_{i,k} + \beta_{i,k} - 2}$. The proximity of the mean to the true probability reflects the average confidence level of the estimated distribution itself, while the mode corresponds to the point of maximum probability density in the distribution. Table II and Table III respectively present the MAE statistics between the mean/mode of the posterior estimates and the true probability for the four baseline schemes mentioned above and the proposed DisBayesCKC framework.

TABLE II
MAE BETWEEN POSTERIOR MEAN AND TRUE PROBABILITY

Paradigm	Barabasi-Albert	Watts-Strogatz	star	ring
W-Obs	0.00590	0.00590	0.00590	0.00590
Cent-CKC	0.00645	0.00645	0.00645	0.00645
Ind-Obs	0.01845	0.01845	0.01845	0.01845
BetaDS-FB	0.01153	0.01101	0.01251	0.01321
DisBayesCKC	0.00637	0.00641	0.00652	0.00624

During the simulation, all topologies alter their specific connection relationships four times, while the graph generation method remains unchanged. Regarding the mean (Table II), our proposed framework achieves a lower MAE than the

TABLE III
MAE BETWEEN POSTERIOR MODE AND TRUE PROBABILITY

Paradigm	Barabasi-Albert	Watts-Strogatz	star	ring
W-Obs	0.00591	0.00591	0.00591	0.00591
Cent-CKC	0.00593	0.00593	0.00593	0.00593
Ind-Obs	0.01846	0.01846	0.01846	0.01846
BetaDS-FB	0.01153	0.01101	0.01251	0.01321
DisBayesCKC	0.00598	0.00600	0.00645	0.00657

agents' estimate under the Centralized-CKC paradigm on three topologies (excluding the star graph). For the mode (Table III), the MAE of our framework is higher than that of Centralized-CKC paradigm on all four topologies. However, the maximum MAE is only 0.00657. While both are distributed coordination frameworks, DisBayesCKC consistently outperforms BetaDS-FB in terms of MAE. This performance gap stems from information granularity: DisBayesCKC preserves the full statistical characteristics of beliefs via LogOP fusion, whereas BetaDS-FB suffers from information loss during the conversion between continuous distributions and discretized evidence. BetaDS-FB's identical mean and mode MAEs stem from using a single Pignistic probability (BetP) scalar as the point estimate for both.

Fig. 7 illustrates the temporal evolution of the estimated means for a triplet under Barabasi-Albert topology. Specifically, Fig. 7(a) compares the aggregated performance of the proposed DisBayesCKC method against the global posterior and the Centralized-CKC baselines, while Figs. 7(b)-(d) detail the individual estimation trajectories of all 10 agents divided into three subgroups. As shown in Fig. 7(a), the curve of the proposed DisBayesCKC aligns closely with both the global posterior and the Centralized-CKC, indicating that the distributed approach effectively approximates the performance of centralized systems. All three curves rapidly converge towards the true probability around 0.74. In contrast to the smoothed aggregated curves in Fig. 7(a), the individual trajectories in Figs. 7(b)-(d) exhibit more noticeable fluctuations during the initial time steps. However, despite these transient oscillations, all agents eventually reach a consensus and consistently converge to the ground truth, demonstrating the robustness and consistency of the proposed algorithm across the network.

B. Sparse Observation Experiment

In the previous experiment, one randomly selected agent observes all triplets at each time step. To further evaluate the performance of DisBayesCKC without global observation, we add a sparse observation experiment. Except for the observation setting, this experiment keeps the system parameters, communication interval, network topologies, baselines, and evaluation metrics the same as those in VII-A. Specifically, at each time step, each agent independently observes only 10% of the triplets in U . The Bayesian update and communication rules follow the sparse implementation in IV-C. That is, the agent only updates the triplets actually observed in the current step, and only sends the posterior parameters of newly observed triplets in the current observation period.

The results are shown in Table IV and Table V. Under sparse observations, the estimation errors of all methods increase, which is expected because each agent can only obtain 10% local observations at each time step and the available information of each single agent is reduced. Nevertheless, DisBayesCKC still keeps a low MAE. For posterior mean estimation, the MAE values of DisBayesCKC under the four topologies are 0.00710, 0.00652, 0.00662, and 0.00714, respectively. These values are clearly better than Ind-Obs with 0.01859, and BetaDS-FB with 0.01425, 0.01269, 0.01773, and 0.01683. Meanwhile, its performance remains close to Cent-CKC. For posterior mode estimation, the MAE values of DisBayesCKC are 0.00670, 0.00622, 0.00618, and 0.00700 under the four topologies, which are also better than Ind-Obs and BetaDS-FB, and close to Cent-CKC.

TABLE IV
MAE BETWEEN POSTERIOR MEAN AND TRUE PROBABILITY (SPARSITY)

Paradigm	Barabasi-Albert	Watts-Strogatz	star	ring
W-Obs	0.00590	0.00590	0.00590	0.00590
Cent-CKC	0.00662	0.00662	0.00662	0.00662
Ind-Obs	0.01859	0.01859	0.01859	0.01859
BetaDS-FB	0.01425	0.01269	0.01773	0.01683
DisBayesCKC	0.00710	0.00652	0.00662	0.00714

TABLE V
MAE BETWEEN POSTERIOR MODE AND TRUE PROBABILITY (SPARSITY)

Paradigm	Barabasi-Albert	Watts-Strogatz	star	ring
W-Obs	0.00591	0.00591	0.00591	0.00591
Cent-CKC	0.00608	0.00608	0.00608	0.00608
Ind-Obs	0.01856	0.01856	0.01856	0.01856
BetaDS-FB	0.01425	0.01269	0.01773	0.01683
DisBayesCKC	0.00670	0.00622	0.00618	0.00700

For the convergence process, Fig. 8(a) shows the evolution of the estimated mean of DisBayesCKC, the global posterior, and Centralized-CKC under sparse observations. Compared with Fig. 7(a) under the full-observation setting, Fig. 8(a) converges more slowly, fluctuates for a longer time, and needs more observation periods before approaching the true probability. This is reasonable because each agent only obtains 10% triplet observations at each time step in the sparse setting. Thus, each round provides less evidence and the posterior distribution needs more observation and communication rounds to converge. Nevertheless, the curve of DisBayesCKC in Fig. 8(a) still approaches the global posterior and Cent-CKC in the later stage, and finally stabilizes near the true probability. Similarly, compared with Fig. 7(b)-(d), the individual estimation trajectories in Fig. 8(b)-(d) are more scattered in the early stage and have longer oscillations. This indicates that sparse observations reduce the early estimation stability of each agent. However, as observation, communication, and fusion proceed, the estimates of different agents gradually approach each other and finally reach consensus.

These results show that DisBayesCKC does not rely on one agent having complete observations of all triplets. Under

sparse local observations, the posterior parameters exchanged among neighbors can compensate for the information shortage of each individual agent, allowing the group to still obtain an accurate and consistent cognitive KB.

C. Triplets Extraction Performance

This section aims to evaluate the effectiveness of the proposed text-to-triplet and triplet-to-text conversion models via simulation experiments. We state how we obtain the EQA language-to-triplets dataset and the model training performance.

1) *Data Set*: The dataset used in our study is a semantic-aligned corpus specifically constructed for indoor EQA scenarios. The dataset is generated by LLMs, specifically ChatGPT and DeepSeek, which are prompted to produce scene-graph-based descriptions for common daily indoor environments. Each natural language sentence is annotated based on a pre-defined cognitive knowledge graph, ensuring that every semantic element—such as scenes, objects, attributes, and states—is consistently aligned with nodes in a unified conceptual space. Through this alignment process, all sentences are converted into standardized semantic triplets, guaranteeing structural uniformity and semantic consistency across the dataset.

We constructed two datasets, one for natural language to triplets, and one for triplets to natural language. The two sets share the identical semantic content, the only difference lies in their input output formats. In the former, natural language sentences serve as inputs and the model outputs the corresponding extracted triplets. In the latter, triplets are fed into the model and the output is a natural language description. The corpus contains both interrogative and declarative sentences, totaling 1,987 samples: 978 questions and 1,009 statements. This dataset includes a total of 4,205 triplets with 706 entities in household indoor scene. The data serve two main purposes: the 1,987 QA pairs are used to fine-tune the T5-Small model, while the related entity data are used to train the Triplet-EmPress compression model. Based on our experimental results, the data volume is sufficient for EQA tasks in simple household scenarios.

Based on the complexity of the EQA sentence, a single sentence can be decomposed into a set of triplets of varying numbers. Table VI shows the statistics of the dataset by classifying data samples by the number of extracted triplets.

TABLE VI
DATASET STATISTICS

Number of Triplets	1	2	3	4	5	6
Number of Samples	582	756	495	146	6	2

2) *Training Performance*: To evaluate the convergence characteristics and stability of the proposed bidirectional semantic extraction model, we track the evolution of both the training and validation losses for the two tasks, language-to-triplets and triplets-to-language, throughout the entire training process, as illustrated in Fig. 9 and Fig. 10.

Fig. 9 illustrates the training loss curves, while Fig. 10 presents the corresponding validation loss curves. The training

loss curves show a rapid decline during the early training stage (0.0–1.0 epochs). Although minor fluctuations appear later in the process, the overall trend continues downward and ultimately converges to a low level. This behavior indicates that the model effectively learns the bidirectional mapping between natural language expressions and semantic triplets and that its optimization process exhibits stable convergence. Similarly, the validation loss decreases sharply during the initial training stage (0.0–2.0 epochs) before gradually stabilizing. After approximately four epochs, the validation loss reaches a steady low value, demonstrating strong generalization capability and robust performance on unseen data.

To demonstrate the effectiveness of the fine-tuning data scale, we added some corresponding experiments, as shown in Fig. 11. The figure presents the curve of loss value changes on the validation dataset as the amount of training dataset increases. Among the 1,987 data entries, 80% (1,589 entries) were used for training, and the remaining were used for validation. As can be seen from the figure, when the amount of training data exceeds 80% of the 1,589 data, the performance improvement tends to stabilize. Moreover, we also test the conversion accuracy for sentence-to-triplets and triplets-to-sentence, as shown in Table VII and Table VIII. It can be observed that under the current data scale, a relatively high conversion accuracy has already been achieved.

TABLE VII
ACCURACY FOR TRIPLET-SENTENCE CONVERSION

Sentence complexity	Total #	Correct #	Accuracy
One triplet	30	30	100.00%
Two triplets	50	49	98.00%
Three triplets	50	45	90.00%
Four triplets	30	26	86.67%
Total	160	150	93.75%

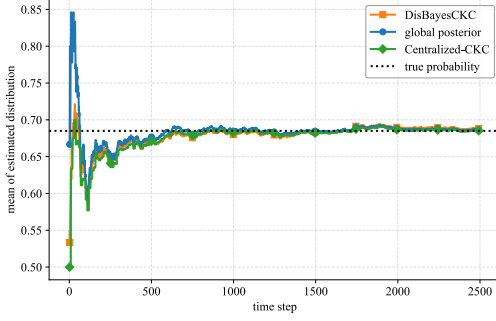
TABLE VIII
ACCURACY FOR SENTENCE-TRIPLET CONVERSION

Sentence complexity	Total #	Correct #	Accuracy
One triplet	30	30	100.00%
Two triplets	50	48	96.00%
Three triplets	50	42	84.00%
Four triplets	30	24	80.00%
Total	160	144	90.00%

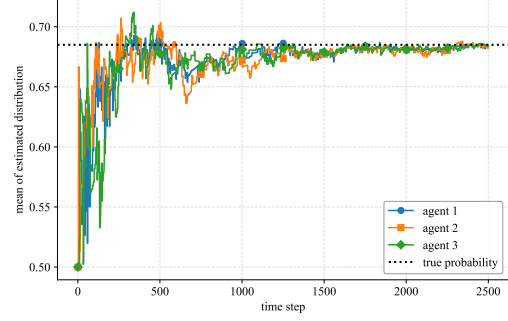
D. Semantic Transmission Performance

1) *Simulation Setting*: We first detail the simulation setting used for evaluating the performance of the semantic transmission.

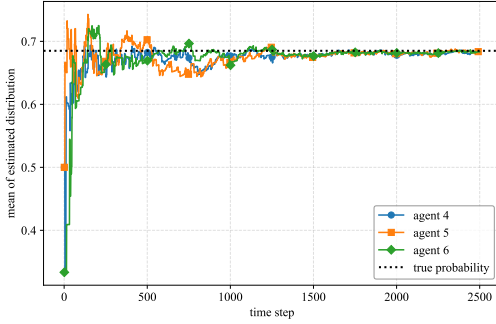
Model and Training Parameters. Both the semantic encoder and decoder adopt an architecture composed of fully connected layers and Highway networks. The key hyperparameter settings used during their training and inference processes are shown in Table IX.



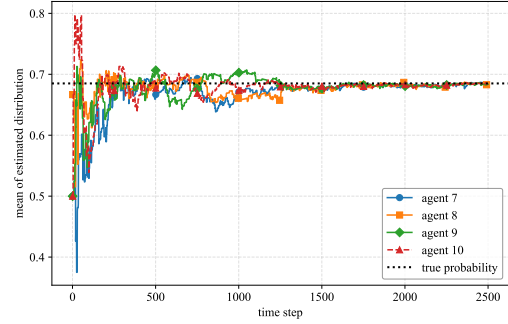
(a) Performance comparison among DisBayesCKC, global posterior, and Centralized-CKC.



(b) Estimation trajectories of Agents 1, 2, and 3.



(c) Estimation trajectories of Agents 4, 5, and 6.



(d) Estimation trajectories of Agents 7, 8, 9, and 10.

Fig. 8. Temporal evolution of the estimated mean distribution under sparse observation, comparing the proposed DisBayesCKC method with baselines and illustrating individual agent convergence.

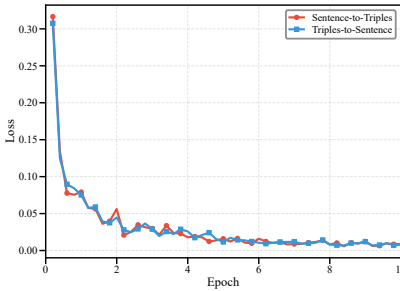


Fig. 9. Training Loss for Bidirectional Semantic Extraction

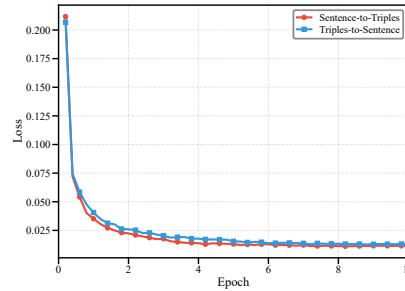


Fig. 10. Validation Loss for Bidirectional Semantic Extraction

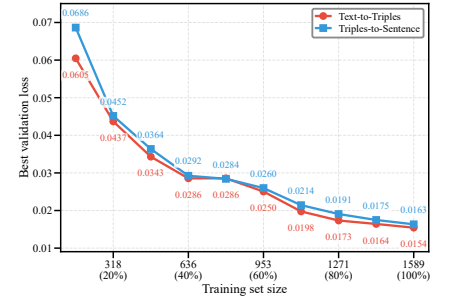


Fig. 11. Convergence with training-size scaling

TABLE IX
BASIC MODEL PARAMETER SETTINGS

Parameter Name	Symbol	Value
Input Dimension	D_{in}	300
Output Dimension	D_{out}	200(Variable)
Epochs	Epochs	1000
Learning Rate	lr	0.01
Quantization Amplitude	A	1
Data Split	Ratio	7:3

Channel and Modulation Parameters. To evaluate the robustness of the model under different channel conditions, AWGN and Rayleigh channels were simulated in the experiments. Transmission tests were conducted under different Signal-to-Noise Ratio (SNR) in the range of (0-25 dB) with an interval of 2 dB. Specific parameters are shown in Table X.

Baseline Algorithms. We compare to the following tra-

ditional communication mechanism and KG triplets oriented semantic communication mechanism.

- **Baseline 1: ASCII (UTF-8) + LDPC Traditional Communication Scheme.** This baseline adopts a traditional layered communication architecture. Entity names are first mapped into fixed-length ASCII bitstreams via UTF-8 encoding, followed by channel coding using an LDPC encoder with a rate of $(512, 5/8)$, and transmitted over the physical channel based on the 16-QAM modulation scheme. This scheme fully adheres to the classic communication link of “source coding–channel coding–modulation”. Serving as a reference system without semantic awareness capabilities, it is utilized to benchmark the performance lower bound of traditional communication in semantic recovery tasks.
- **Baseline 2: KG-SC Semantic Communication Scheme [20].** KG-SC is a semantic communication

TABLE X
EXPERIMENTAL CONFIGURATIONS OF SEMANTIC COMMUNICATION AND TRADITIONAL BASELINES

Method	Source Coding	Channel Coding	Modulation	Neural Architecture
ASCII(UTF-8)+ LDPC	ASCII(UTF-8)	LDPC (512, $\frac{5}{8}$, BP)	16-QAM	–
KG-SC	–	–	16-QAM	Dense + Pooling + linear(binary quant).
Triplet- EmPress(Ours)	–	–	16-QAM	FCN + Highway + STE binary quant.

method based on knowledge graph embeddings. It performs continuous semantic compression on entity embeddings through a “Dense + Pooling + Linear” structure and implements information transmission using binarization quantization, with the modulation scheme also being 16-QAM. Compared with traditional communication, this scheme utilizes neural networks to accomplish joint semantic compression and anti-noise mapping, representing a typical deep semantic communication baseline method.

Evaluation Metrics. The performance of the compared communication mechanisms is evaluated through the following evaluation metrics.

(1) **Fidelity** measures the signal quality of the reconstructed vectors:

$$F_i = 1 - \frac{\|T_i - \hat{T}_i\|_2^2}{\|T_i\|_2^2} \quad (25)$$

(2) **Recall@k** measures the semantic preservation capability. This study adopts Euclidean distance to retrieve similar entities, which is a common method for determining semantic consistency in knowledge graph embeddings. For the reconstructed vector $\hat{T}_i$, the k nearest original vectors in the test set are retrieved. If the ground truth T_i is among them, it is considered a hit:

$$\text{Recall@k} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(\text{rank}_i \leq k) \quad (26)$$

where $\mathbf{1}(\cdot)$ is the indicator function, which equals 1 if the condition holds and 0 otherwise; a smaller rank_i indicates a higher ranking position.

(3) **MRR** reflects the continuous variation of the ranking:

$$\text{MRR} = \frac{1}{N} \sum_{i=1}^N \frac{1}{\text{rank}_i} \quad (27)$$

2) *Data Compression:* We conduct a quantitative comparison of the number of transmitted bits associated with transmitting natural-language sentences versus structured encoded triplets. We compare Triplet-EmPress with ASCII+LDPC. The later first encode characters with ASCII, then with LDPC encoder of rate (512, 5/8). For the proposed Triplet-EmPress mechanism, if multiple triplets are extracted from a single sentence, the triplets not contained in the cognitive KB are transmitted through the proposed Triplet-EmPress encoding method, while the known triplets are substituted by their

corresponding indices. Each index occupies 128 bits, whereas the encoded triplets requires 300 bits for transmission.

Based on these encoding rules and the characteristics of the evaluated dataset, we compute the average bit overhead for transmitting a full sentence and a single entity under the two schemes. The detailed comparison is summarized in Table XI. The results are calculated from the 1987 samples from the EQA dataset.

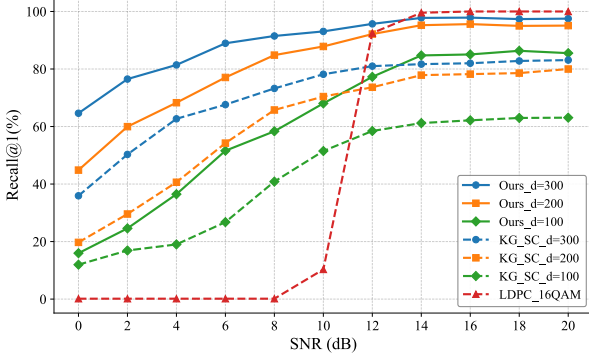
TABLE XI
BIT COST COMPARISON

Method	One EQA Sentence (bits)	One Triplet Entity (bits)
ASCII + LDPC	4743.362	512
Triplet-EmPress	442.854	100 / 200 / 300

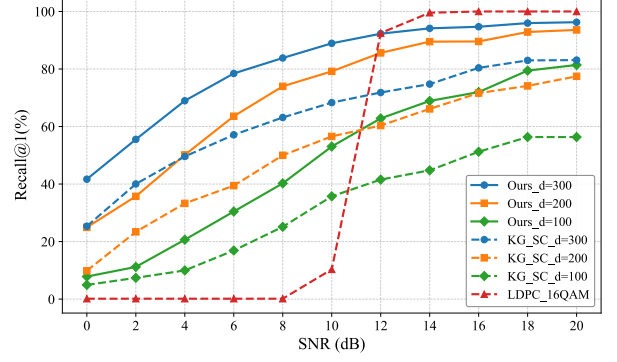
3) *Transmission Performance:* Finally, we show the transmission performance of the proposed Triplet-EmPress mechanism under different transmission conditions, comparing to the baseline methods. The results are reported in terms of both retrieval effectiveness and representation quality over a wide range of signal-to-noise ratios (SNRs). Specifically, Fig. 12 and Fig. 13 present the Recall@1 and Recall@5 performance under AWGN and Rayleigh fading channels, highlighting the robustness of the proposed method in retrieval-oriented tasks. In addition, Fig. 14 and Fig. 15 illustrate the reconstructed semantic fidelity and MRR, which jointly characterize semantic preservation and ranking stability under channel impairments.

From these results, we first observe that the semantic communication mechanism exhibits dimensionality sensitivity, with high-dimensional compressions possessing advantage in combating noise. Taking the AWGN channel as an example, in the low SNR region (SNR = 0 dB), the system employing $d = 300$ dimensional compression achieves an accuracy of approximately 65% on the Recall@1 task, representing an improvement of nearly 50% compared to the low-dimensional model with $d = 100$ (approximately 15%). Furthermore, the leniency of the task yields a significant gain in the hit rate. Under identical channel conditions, the Recall@5 metric improves by an average of approximately 20% compared to Recall@1, indicating that even if the model fails to achieve an exact hit at the top position, it retains the target semantics in the forefront of the candidate list with high probability.

A comparison of performance under different channel conditions reveals that the challenges posed by the Rayleigh fading channel to semantic recovery are primarily concentrated in the low SNR interval. The results indicate that when SNR < 6 dB, the MRR and Fidelity metrics under the Rayleigh

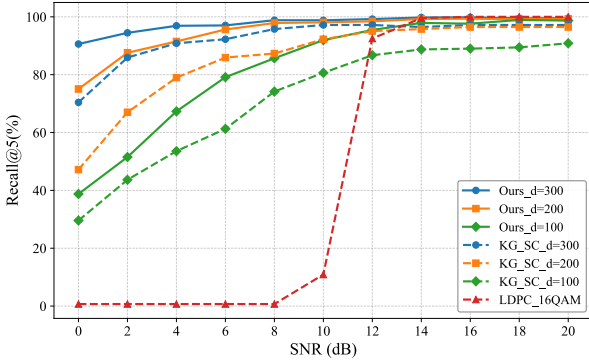


(a) Recall@1 (AWGN)

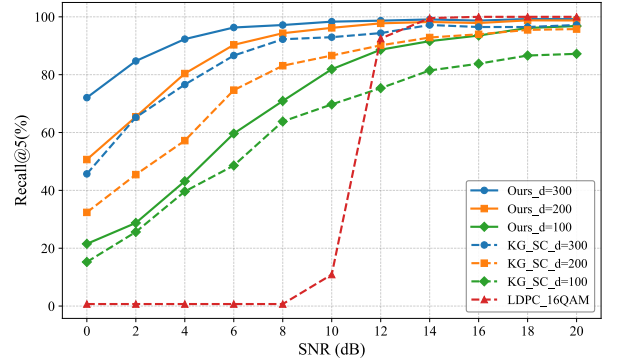


(b) Recall@1 (Rayleigh)

Fig. 12. Recall@1 performance under different SNR. Recall@1 under AWGN, Recall@1 under Rayleigh fading channels.

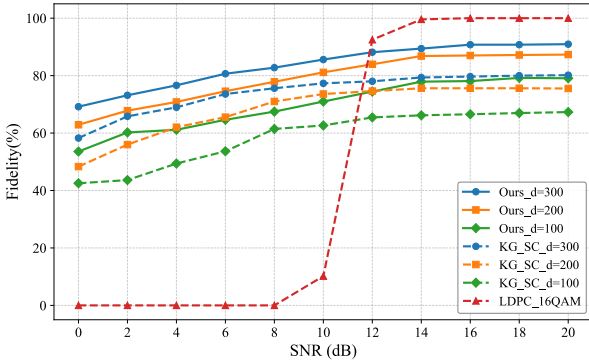


(a) Recall@5 (AWGN)

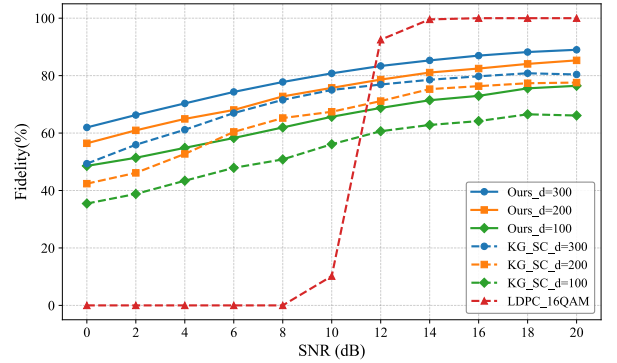


(b) Recall@5 (Rayleigh)

Fig. 13. Recall@5 performance under different SNR. Recall@5 under AWGN, and Recall@5 under Rayleigh fading channels.



(a) Fidelity (AWGN)



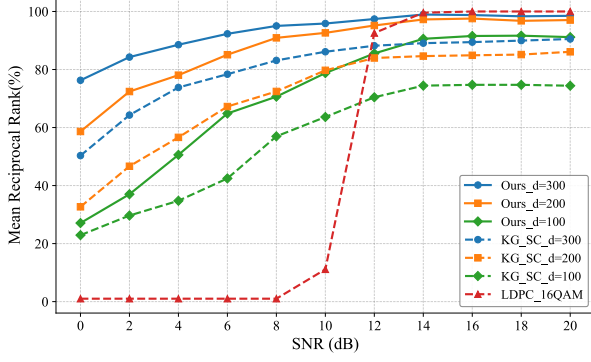
(b) Fidelity (Rayleigh)

Fig. 14. Reconstruction under different SNR. Fidelity under AWGN and Fidelity under Rayleigh fading channels.

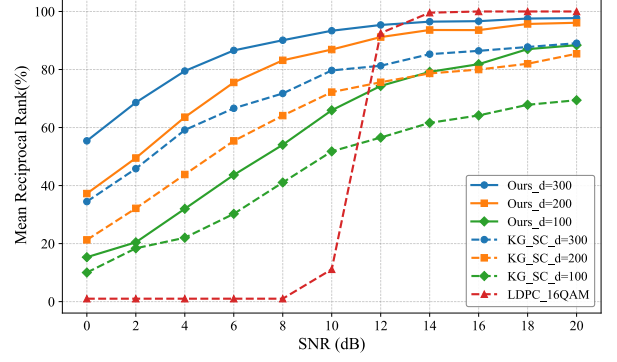
channel are on average approximately 8% to 20% lower than those under the AWGN channel. However, the proposed Triplet-EmPress model demonstrates good robustness, as the SNR increases to above 10 dB, the two curves gradually converge, eventually both reaching a performance of over 90%. This confirms that even in complex channels with multipath fading, as long as the SNR reaches a moderate level, the proposed scheme can achieve high-quality semantic

transmission approximating that of the Gaussian channel.

Comparison with existing baseline schemes further verifies the superiority of Triplet-EmPress. The traditional communication system (LDPC_16QAM) exhibits a “cliff effect”: when the SNR is below 6 dB, due to the bit error rate exceeding the error correction limit, its Recall@1 and MRR metrics rapidly drop to 0%, resulting in a complete loss of semantic recovery capability. In contrast, Triplet-EmPress maintains effective



(a) MRR (AWGN)



(b) MRR (Rayleigh)

Fig. 15. Ranking robustness under different SNR. MRR under AWGN and MRR under Rayleigh fading channels.

semantic discrimination within this “blind zone”. Meanwhile, benefiting from the extracting of key features by the Highway Gating mechanism, the semantic fidelity of this mechanism under low SNR is approximately 10% higher than that of the KG-SC scheme which employs a fixed compression structure, demonstrating the effectiveness of dimension-wise semantic path selection under strong noise interference.

In summary, the experimental results fully demonstrate that the Triplet-EmPress system significantly outperforms the traditional LDPC scheme and the KG-SC baseline in terms of low-level semantic recovery, ranking consistency, and retrieval tasks, particularly in the low SNR region (0~6 dB) where traditional communication fails. The results demonstrate the effectiveness of Triplet-EmPress mechanism for transmitting the semantics of triplets, which is then utilized for agent collaboration downstream tasks.

E. EQA Task Performance

We also performed a set of simulations to evaluate the performance of proposed mechanisms in EQA task. We use the Habitat Matterport 3D Research Dataset (HM3D), a widely used simulation platform for embodied AI, as the execution environment for the EQA task [41]. Then, we build a complex embodied question answering dataset based on HM3D as shown in Fig. 16. We selected a home environment as shown in Fig. 16 and constructed the corresponding hierarchical scene graph for this environment. We then generated 150 EQA question-answer pairs for this environment using GPT-5.4. These question-answer pairs are all focused on querying about a specific object in the environment. Each question can be parsed into several triplets, where the first $n - 1$ triplets are used for locating and navigating to the object in the environment, and the final triplet asks about the object’s state, attribute, positional relationship, and so on. Then, the question and the scene taken from HM3D are feed into the All-Seeing Model V2 to obtain the answer.

The simulation process of the EQA task is shown in Fig. 17. The input query Q is first extracted into triplets form Q' . After compression, transmission and decompression, the triplets Q'' is obtained on the remote agent. The remote agent uses Q''

to locate the object, navigate, and observe the scene, forming a query to obtain the initial answer A_1 from the All-Seeing Model V2. Then, this answer A_1 is extracted into triplets A'_1 and transmitted to the personal agent, yielding A'_1 and the final answer A_2 . During this process, the final task accuracy is affected by triplet extraction, compression and transmission, and the scene reasoning capability of the All-Seeing Model V2. Therefore, we account for multiple accuracy metrics based on the results.

The experimental results are presented in the Table XII. Within the process shown in Fig. 17, the accuracy of $Q-Q'$ refers to the correctness rate of triple conversion for the 150 EQA questions in the end-to-end experiment, which is 94.67%. The accuracy of $Q''-A_1$ is mainly affected by the image reasoning and understanding capability of the All-Seeing Model V2, with an accuracy of 86%. Based on the analysis of incorrect A_1 , the main errors are concentrated on complex spatial relationship understanding and object recognition errors. The accuracy of the remaining stages is primarily influenced by communication transmission. We measured the performance of baseline methods under different communication conditions. As shown in Table XII, compared to the entity transmission performance in the previous sections, the overall accuracy decreases because each triple contains three entities, and correctly parsing the original QA requires correctly retrieving all three entities. However, under low signal-to-noise ratio conditions, our proposed method still maintains a higher end-to-end accuracy compared to other methods.

VIII. CONCLUSION

This paper presents a comprehensive agentic semantic communication framework for collaborative EQA multi-agent systems. By integrating the agent’s cognitive KB into semantic exchange, we establish a foundational paradigm that shifts communication from raw data to semantic information. The proposed distributed coordination mechanism ensures group-wide cognitive consistency, while the tailored semantic extraction and encoding methods significantly enhance transmission efficiency. This work provides essential solutions for efficient agentic communication in intelligent multi-agent networks.



Fig. 16. The simulation environment and the EQA data set.

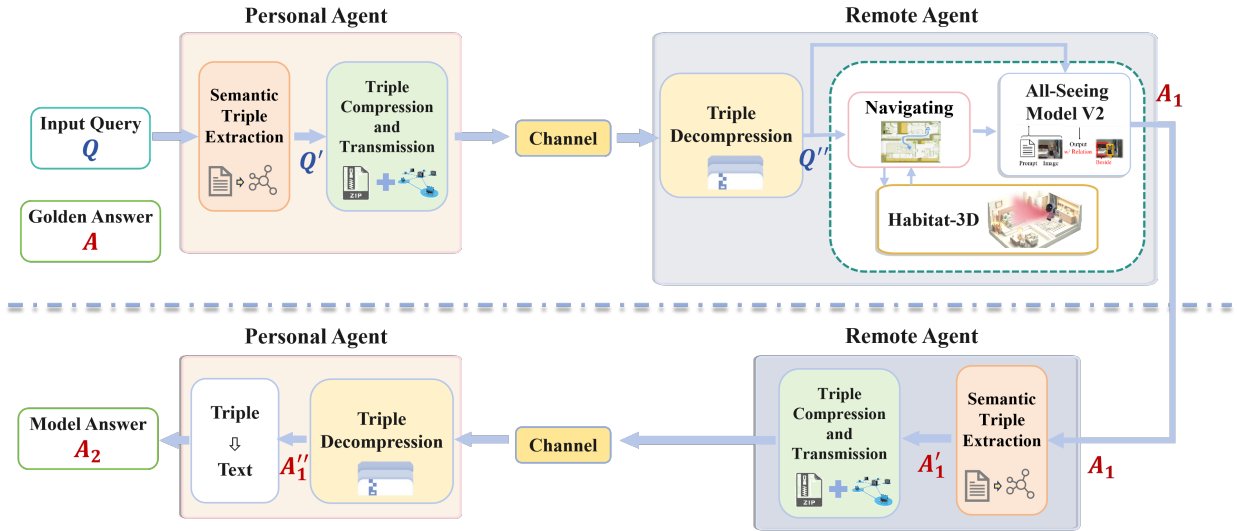


Fig. 17. The EQA task evaluation process.

For future works, we consider the following challenges: practical deployment in dynamic and unstructured environments, robustness of the cognitive consensus mechanism under non-ideal perception, adapting the framework for lightweight IoT devices, lifelong learning for dynamic knowledge base updates, and scalable validation in large-scale heterogeneous multi-agent systems.

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TABLE XII
DIFFERENT LEVEL OF ACCURACY FOR E2E EQA TASK (%)

Methods	$Q'-Q''$			A1-A2			A-A2		
	6dB	8dB	10dB	6dB	8dB	10dB	6dB	8dB	10dB
LDPC	0.71	1.24	98.21	0.67	1.33	92.67	0.00	0.00	78.67
Ours d=300	60.31	62.45	69.15	56.67	58.67	65.33	29.33	32.00	38.67
Ours d=200	46.88	47.20	53.94	44.00	44.67	50.67	18.00	18.00	23.33
Ours d=100	24.20	24.51	27.91	22.67	23.33	26.00	4.67	4.67	6.67
KG_SC d=300	30.45	30.85	35.83	28.67	29.33	34.00	7.33	8.00	10.67
KG_SC d=200	26.90	28.27	30.94	25.33	26.67	29.33	6.00	6.67	8.00
KG_SC d=100	10.84	13.05	16.30	10.00	12.00	15.33	0.67	1.33	2.00

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